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Constrained Belief Updates Explain Geometric Structures in Transformer Representations

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Abstract

What computational structures emerge in transformers trained on next-token prediction? In this work, we provide evidence that transformers implement constrained Bayesian belief updating—a parallelized version of partial Bayesian inference shaped by architectural constraints. We integrate the model-agnostic theory of optimal prediction with mechanistic interpretability to analyze transformers trained on a tractable family of hidden Markov models that generate rich geometric patterns in neural activations. Our primary analysis focuses on single-layer transformers, revealing how the first attention layer implements these constrained updates, with extensions to multi-layer architectures demonstrating how subsequent layers refine these representations. We find that attention carries out an algorithm with a natural interpretation in the probability simplex, and create representations with distinctive geometric structure. We show how both the algorithmic behavior and the underlying geometry of these representations can be theoretically predicted in detail—including the attention pattern, OV-vectors, and embedding vectors—by modifying the equations for optimal future token predictions to account for the architectural constraints of attention. Our approach provides a principled lens on how architectural constraints shape the implementation of optimal prediction, revealing why transformers develop specific intermediate geometric structures.

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1. Introduction

Transformers excel at next-token prediction (Vaswani et al., 2017), but their success belies a fundamental tension: optimal prediction requires Bayesian belief updating, a recursive process, while their architecture enforces parallelized, attention-driven computation (Fig. A1). How do transformers resolve this conflict? We show that they develop geometrically structured representations that approximate Bayesian inference under architectural constraints, revealing a precise interplay between theoretical necessity and implementation.

In this work, we combine insights from the theory of optimal prediction with neural network analysis. First, computational mechanics (Shalizi & Crutchfield, 2001a; Marzen & Crutchfield, 2017; Riechers & Crutchfield, 2018b; Pepper, 2024; Shai et al., 2024) dictates *what* an optimal predictor must represent: belief states that encode distributions over futures. Second, mechanistic interpretability reveals *how* transformers approximate these states under architectural constraints, bending Bayesian updates into attention's parallelizable form (Elhage et al., 2021; Nanda et al., 2023).

By combining these frameworks we reveal *why* transformers learn certain intermediate structures. We find that the geometry of a transformer's internal representations is not an accident—it is a mathematical signature of how architectural constraints warp otherwise optimal Bayesian inference. By interpreting learned weights and activations via standard mechanistic interpretability, we uncover an algorithm that is well-captured by the **constrained belief updating** equations. From first principles, we derive the constrained belief geometries, and reverse-engineer the transformer's computational blueprint, predicting attention patterns, value vectors, and residual stream geometries precisely. Thus, beyond verifying that transformers encode belief states, we show how the specific circuits that implement those states necessarily deviate from the unconstrained Bayesian ideal in predictable and theoretically tractable ways.

To concretize these ideas, we focus on transformers trained on data from the Mess3 class of hidden Markov models (HMMs) (Marzen & Crutchfield, 2017), which provides rich and visualizable belief-state geometries and also admits a tractable optimal predictor. Our primary analysis exam-

ines single-layer transformers to isolate how the first attention layer implements constrained belief updating, though we also demonstrate that these principles extend to multi-layer architectures where subsequent layers refine the initial constrained representations. This allows us to rigorously compare the theoretically optimal geometry with the neural-activation geometry that transformers learn. More broadly, we anticipate that the same tension between architecture and optimal inference arises in large language models trained on natural text, and that our methodology would shed light on those more complex cases.

Key contributions:

1. **A Unified View of Optimal Prediction and Transformer Computation**: We bridge the model-agnostic theory of Bayesian belief states with the model-specific constraints of attention-based parallel processing. This synthesis explains why transformers trained on next-token prediction discover a distinct "constrained belief updating" geometry—balancing optimal Bayesian inference with the functional form of attention.

2. **Spectral Theory of Constrained Belief Updating**: We develop a theoretical framework that analyzes how eigenvalues of the data-generating transition matrices determine attention heads' behavior. By decomposing belief updates spectrally, we show that multi-head attention naturally implements these scalar updates in orthogonal modes—even handling oscillatory decay of influence—through a sum of specialized head outputs.

3. **Predictive Experiments and Mechanistic Verification**: Our approach yields specific, testable predictions about attention patterns, value vectors, intermediate fractal representations, and final belief-state geometry. We confirm these predictions in trained transformers, demonstrating how the inherently recurrent next-token task is realized by an attention-based, parallelized implementation of Bayesian belief updates.

2. Background

2.1. Related Work

Features as directions in activation space Modern interpretability research views neural network representations through the lens of linear geometry, analyzing how activation patterns align with specific directions that encode fundamental features (Park et al., 2024). This perspective is particularly useful given superposition (Elhage et al., 2022), where networks encode more features than available neurons using non-orthogonal vectors. Conceptualizing features as linear directions has been instrumental (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024) in understanding what information transformers represent, with geometric relationships between features revealing structured internal representations (Engels et al., 2024). Our work provides a mechanistic explanation for these non-orthogonal geometric structures, providing the theoretical *why* to complement the *what* of feature representations.

From features to circuits While feature directions reveal what information is encoded, understanding how networks process this information is done by identifying computational circuits—subnetworks that implement specific algorithmic operations. Circuits typically combine simpler features into more complex ones as information flows through the network. Examples include circuits that detect syntax (Elhage et al., 2021), implement indirect object identification (Wang et al., 2022), or perform basic arithmetic (Nanda et al., 2023). However, identifying circuits remains largely a manual process, starting from observed behaviors and working backwards to discover relevant components (although active research is developing automated approaches; see Conmy et al. (2023); Marks et al. (2024)).

Our work demonstrates that a principled, top-down theoretical framework, based on constrained belief updating, can guide the search for circuits and provide a deeper understanding of their function within the larger network. We show how specific circuits in the attention mechanism directly implement the computations predicted by our theory.

Belief state geometry and computational mechanics Our work draws inspiration from computational mechanics, a framework for studying information processing in dynamical systems (Shalizi & Crutchfield, 2001b; Crutchfield, 2012; Riechers & Crutchfield, 2018b). When applied to sequential data, computational mechanics, in accordance with the POMDP framework (Kaelbling et al., 1998), shows that optimal prediction requires maintaining beliefs about the underlying latent states of the data-generating process (Upper, 1997). These belief states can be visualized as points on a probability simplex, evolving according to Bayesian updating rules, and forming characteristic geometric patterns (Crutchfield, 1994; Marzen & Crutchfield, 2017). Recent work shows that transformers naturally discover and encode these belief state geometries in their activations (Shai et al., 2024). This connection offers a principled way to analyze network representations: rather than reverse-engineering observed behaviors, we can study how architectural constraints shape the network's implementation of theoretically optimal prediction strategies.

This is the approach taken here. We move beyond prior work by proposing and validating a theory of constrained belief updating, demonstrating how specific architectural elements, like the attention mechanism, modify the idealized belief state dynamics. This perspective shifts the focus from

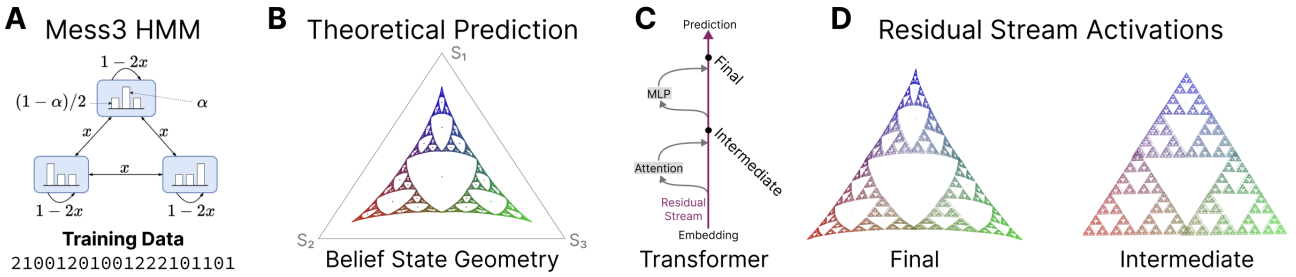

Figure 1: Transformers' internal representations exhibit complex geometric structure matching the belief-state geometry. **(A)** Mess3 HMM, vertices represent hidden states with their emission distributions. **(B)** Ground-truth belief state geometry of Mess3. Each point represents a belief-state probability distribution over hidden states of the HMM, induced via Bayesian updates upon a sequence of observed emissions, with proximity to the vertices of the simplex corresponding to the probabilities of the three hidden states. **(C)** Schematic of a single-layer transformer with Intermediate activations after Attention, and Final activations after the subsequent MLP. **(D)** PCA projections of the model's final residual stream (left), before the unembedding, reveals a geometric representation that closely matches the belief geometry shown in (B), whereas the PCA projection of the intermediate residual stream (right) after attention but before the MLP exhibits an intricate but different structure. In (B) and (D), points are colored according to the ground-truth belief states associated with the sequence of tokens that induces the point, taking the three constituent probabilities over hidden states of the HMM as RGB values.

reverse-engineering learned features to understanding why particular geometric patterns emerge during training as a consequence of the interplay between optimal prediction and architectural constraints. Our work provides a concrete example of how this theoretical framework can be applied to understand the internal mechanisms of transformers.

2.2. Optimal Prediction and Belief State Geometry

Shai et al. (2024) showed that transformers minimizing next-token loss internally represent the context-induced probability density over the entire future of possible token sequences:

$$\Pr(Z_{\mathrm{d}+1:\infty} | Z_{1:\mathrm{d}} = z_{1:\mathrm{d}}) \tag{1}$$

where $Z_{\mathrm{d}+1:\infty} = Z_{\mathrm{d}+1}, Z_{\mathrm{d}+2}, \ldots$ denotes the sequence of random variables for future tokens, $Z_{1:\mathrm{d}} = Z_1, \ldots, Z_{\mathrm{d}}$ denotes the sequence of random variables for past tokens, which is realized by a particular sequence of tokens $z_{1:\mathrm{d}} \in \mathcal{Z}^{\mathrm{d}}$ known as the context up to position s.

When we conceptualize the training data as being generated by an edge-emitting hidden Markov model (Mealy HMM), we can derive a natural geometric embedding for these conditional probability distributions. HMMs generate training data by emitting tokens when moving among its hidden states $\mathcal{S}$, from one hidden state S_t at time t to the next. The natural geometric embedding is then given by considering how an initial distribution over hidden states $S_0 \sim \boldsymbol{\eta}_{\varnothing}$, as a point in the vector space $\mathbb{R}^{|\mathcal{S}|}$ (with coordinates given by the probability elements), evolves upon seeing a particular sequence of tokens, $z_{1:\mathrm{d}}$. This distribution over the hidden states, which uniquely induces a probability density over all possible futures, is updated via Bayes rule according to the substochastic transition matrices of the HMM, $\left(T^{(z)}\right)_{z \in \mathcal{Z}}$, with matrix elements $T^{(z)}_{s,s'} = \Pr(Z_{t+1} = z, S_{t+1} = s' | S_t = s)$. In particular, the updated distribution, given context $z_{1:\mathrm{d}}$, is

$$\boldsymbol{\eta}_{\varnothing} \mapsto \vec{r}^{(z_{1:\mathrm{d}})}_{\text{full}} = \frac{\boldsymbol{\eta}_{\varnothing} T^{(z_{1:\mathrm{d}})}}{\boldsymbol{\eta}_{\varnothing} T^{(z_{1:\mathrm{d}})} \mathbf{1}}\,, \tag{2}$$

where $T^{(z_{1:L})} = T^{(z_1)} \cdots T^{(z_L)}$, and $\mathbf{1}$ is the column vector of all ones. In this paper, we will make the simplifying assumption that the training data is sampled from a stationary stochastic process, in which case the initial distribution over latent states is the stationary distribution $\boldsymbol{\eta}_{\varnothing} = \boldsymbol{\pi} = \boldsymbol{\pi} T$, where $T = \sum_{z \in \mathcal{Z}} T^{(z)}$ is the row-stochastic transition matrix over hidden states.

Thus, Eq. (2) embeds each token sequence into a probability simplex over the latent states of the HMM—a point in a real-valued vector space. The totality of these points forms a particular geometry, called the belief state geometry, and is universally found in linear form within the activations of various deep neural networks, including RNNs (Pepper, 2024) and transformers (Shai et al., 2024).

This precise framework for anticipating intermediate activations in transformers provides a natural interpretation of the attention mechanism in which it moves information in a belief simplex for the purposes of building up the architecture-independent belief state geometry given in Eq. (2).

3. Methodology

Data Generation. Our study focuses on the Mess3 parametrized family of hidden Markov models (Marzen